Link-Layer Addressing & Address Resolution Protocol (ARP)

Coverage: 48-Bit MAC Addresses (EUI-48), ARP Request (Broadcast) & ARP Reply (Unicast), ARP Cache & Timeout, Subnet Crossing

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1. MAC Addressing & ARP

- 48-bit hardware MAC burned into NIC ROM. ARP resolves IP to MAC on local link.
- ARP Request is L2 Broadcast (FF:..:FF); ARP Reply is L2 Unicast.

2. Subnet Crossing

- L3 IP addresses remain unchanged end-to-end; L2 MAC addresses rewritten at every router hop.

■ KTU University High-Yield Exam Questions

- Q: Explain ARP protocol operation and trace MAC/IP rewriting across subnets. (8 Marks)